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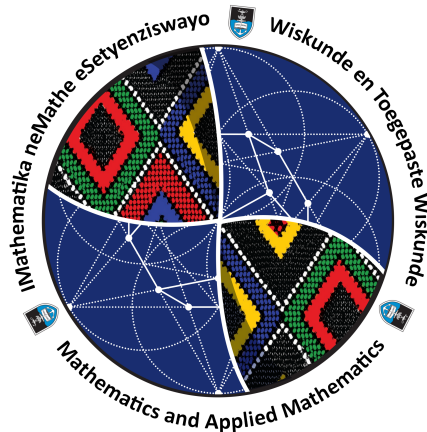
Department of Mathematics and Applied Mathematics
MAM3000W 3DM: Discrete Mathematics

Test 2

15 May 2024

Time : 45 minutes

40 marks = 100%



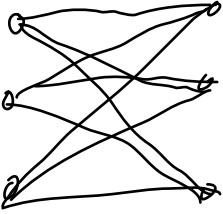
- This question paper consists of **8 pages** (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. The work in your *UCT Examination Answer Book* will not be marked. Only the answers that you write on this question paper will be marked.
- Calculators are not allowed. When your answer contains constants such as π , e or $\sqrt{2}$, leave them in that form. Don't simplify your answers.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.
- For Question A, write the answer only. If you show working you will be penalized.
- For Question B you must prove everything. No proof = no marks, even if your answer is correct.

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Question A. [22 points]

1. Draw an example of each of the following (if such a graph exists) or write Does Not Exist (if it doesn't). No proofs or working necessary.

(a) A nonplanar graph of order $n \geq 3$ and size $m \leq 3n - 6$. (3)

$K_{3,3} :$


✓✓✓

(b) A tree with center P_3 . (3)

Does Not Exist

✓✓✓

(The center of a tree is always K_1
or K_2)

(c) A connected graph in which every vertex is a cut-vertex. (3)

Does not exist.

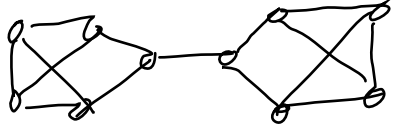
✓✓✓

(HW question)

(d) A cubic graph that contains a bridge.

(3)

There are infinitely many such graphs.
 For example:



✓✓✓

(e) A planar graph G with $\delta(G) = 6$.

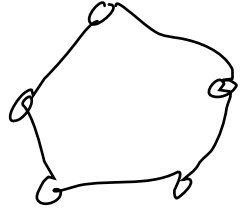
(3)

Does Not Exist. ✓✓✓
 $(G \text{ planar} \Rightarrow \delta(G) \leq 5)$

(f) A graph with chromatic number 3 that does not contain K_3 as a subgraph.

(3)

There are infinitely many such graphs
 For example: C_5



✓✓✓

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2. Define the following terms:

(a) Block.

(2)

Either: A connected graph that contains
no cut-vertices,
or: A maximal connected subgraph that
contains no cut-vertex.



(b) Tree.

(2)

A graph that is connected and
acyclic.



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Question B. [25 points]

For Question B, show all working/proofs! Zero working/proofs = zero marks!
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1. Prove that a graph G of order at least 3 is a block if and only if for every pair u, v of vertices of G , there is a cycle containing both u and v . (10)

This is theorem 13 on p 41 of the nks.

$(\Leftarrow) : 2$

$(\Rightarrow) : 8$

2. Use a double-counting argument to prove that for every nonnegative integer n ,

if not, zero!

$$\sum_{k=0}^n \binom{n}{k} = 2^n.$$

(5)

If $n = 0$, then

$$\sum_{k=0}^0 \binom{0}{k} = \binom{0}{0} = 1 = 2^0,$$

so we assume $n \geq 1$. let N = the number of subsets of $S = \{1, 2, \dots, n\}$. Then:

① Since for each subset, we must choose to include or exclude each of the n elements, the total number of subsets is 2^n , i.e., $N = 2^n$.

② The number of k -subsets of S is $\binom{n}{k}$, so the total number of subsets of all sizes

is

$$N = \sum_{k=0}^n \binom{n}{k}$$

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Thus

$$\sum_{k=0}^n \binom{n}{k} = N = 2^n$$

QED

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3. Suppose that T is a tree of order $n \geq 4$ that contains only vertices of degree 1 and 3. Prove that T contains $(n-2)/2$ vertices of degree 3. (5)

Suppose T has n_1 vertices of degree 1 and n_3 vertices of degree 3.

Then $n_1 + n_3 = n$. Since T is a tree, by a theorem, $m = n - 1$. Thus, by another theorem,

$$2m = \sum_{v \in V(T)} \deg v$$

$$\begin{aligned} \therefore 2(n-1) &= 1 \cdot n_1 + 3 \cdot n_3 \\ &= n - n_3 + 3n_3 \end{aligned}$$

$$\therefore 2n - 2 = n + 2n_3$$

$$\therefore n_3 = \frac{n-2}{2}$$

(Q.E.D.)

4. Prove that the hypercube Q_t is nonplanar for $t \geq 4$.

(5)

Suppose to the contrary, that for some $t \geq 4$, Q_t is planar. By a HW question, Q_t is bipartite. Thus, by a theorem Q_t contains no triangles, so by a discussion in class / a HW question, $m \leq 2n - 4$, where m and n are the size and order of Q_t . By another HW question,

$$n = 2^t$$

$$m = t 2^{t-1}$$

$$\text{Thus } t 2^{t-1} \leq 2^{t+1} - 4$$

$$\Leftrightarrow 2^{t+1} - t 2^{t-1} \geq 4$$

$$\Leftrightarrow (4 - t) 2^{t-1} \geq 4$$

But $t \geq 4 \Rightarrow (4 - t) 2^{t-1} \leq 0$, a contradiction.

Thus $t \geq 4 \Rightarrow Q_t$ is nonplanar.

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